Client Handoff Guide

GSTD Modeling Capstone Deliverables

Purpose: This guide explains how the handoff package is organized and how the client should interpret the final modeling deliverables. It is intended as a practical orientation document, not as a replacement for the final report or technical notebooks.

1. Handoff Package Structure

The package is organized so that the client can start with the high-level materials, then move into data, final notebooks, demo materials, and appendix work as needed.

Folder	Purpose	Primary Audience
00_README_START_HERE	Orientation files that explain what to open first, how the package is organized, and how to run the notebooks.	Client, instructor, future project team
01_Final_Report_and_Executive_Summary	Final written report, executive summary, and presentation materials.	Client and academic reviewers
02_Data	Final validated dataset, raw DSSAT outputs, and short notes on dataset origin and derived modeling columns.	Client technical team
03_Final_Notebooks	Locked technical notebooks for the two final modeling tasks: Task A and Task B.	Technical reviewers and future analysts
04_Model_Results_and_Figures	Exported result tables and model output artifacts for review without rerunning notebooks.	Anyone reviewing outputs
05_Client_Demo_Materials	Polished demo notebooks and HTML exports designed for walkthroughs and client presentations.	Client and presentation audience
06_Appendix_Research	Dataset creation workflow, parser notebooks, preliminary modeling, and supporting research trail.	Future analysts who want background context

2. Recommended Review Order

- Start with README_START_HERE.pdf for the fastest overview of the project, folder structure, and final recommendations.
- Read the Executive Summary and Final Report in 01_Final_Report_and_Executive_Summary.
- Review TaskA_Current_GSTD_Classification_FINAL.ipynb and TaskB_7Day_GSTD5_Alert_FINAL.ipynb as the source-of-truth notebooks for final model results.
- Use the client demo notebooks and HTML exports for walkthroughs with non-technical stakeholders.
- Use Dataset_Creation_and_Validation_Workflow.ipynb if the client wants to verify how the final validated CSV was created from raw DSSAT outputs.
- Use other appendix notebooks only for background context.

3. Final Deliverables and Interpretation

Task A: Current GSTD Classification

Purpose: Classify the current crop growth stage (GSTD) using the final validated dataset.

Final interpretation: The final Task A work shows that current GSTD classification is feasible, especially when client-feasible variables are combined with LAID and past-looking time-series features. Broader/full-signal models act as upper-bound benchmarks, while client-feasible models are more realistic for future use.

Task B: 7-Day GSTD 5 Alert

Purpose: Predict whether a Stage 1 observation is likely to enter GSTD 5 within the next 7 days.

Final interpretation: The final Task B recommendation is the Client-Feasible TS LightGBM model. The optional Client-6 + LAID + TS model performs slightly better in some metrics, but the gain is small, so LAID should be treated as an optional extension if it is reliably available.

Threshold interpretation: The final LightGBM threshold section in the core Task B notebook and the Task B demo notebook should be used for alert-threshold interpretation. The recommended Client-Feasible TS LightGBM Best F1 threshold in the finalized run is 0.9984.

4. Dataset Guidance

The final notebooks assume the cleaned modeling file is named final_validated_dataset.csv. The data folder also includes dataset_notes and derived_columns_notes to explain how the dataset was produced and how the modeling-specific target/features are used.

- Raw DSSAT outputs were parsed from PlantGro.OUT, SoilNi.OUT, SoilWat.OUT, and Weather.OUT.
- The step-by-step parsing, cleaning, merging, validation, and saving workflow is documented in 06_Appendix_Research/Dataset_Creation_and_Validation_Workflow.ipynb.
- DSSATout2Dataframe.ipynb is also included as a parser reference.
- RUNNO is used for grouped validation to reduce leakage across repeated observations from the same simulation run.
- Forward-time splits are used to evaluate future-year robustness.
- Treatment holdout checks are used to test whether models generalize across unseen treatments.

5. Important Limitations

- The dataset is DSSAT/simulation-derived, so final production deployment should include validation on real operational or field data.
- Task B should be interpreted as a decision-support alert, not as an automatic agronomic decision system.
- Operating thresholds are business decisions. A lower threshold catches more true transitions but creates more false alerts; a higher threshold reduces false alerts but may miss more true transitions.
- Earlier exploratory notebooks may contain results from preliminary datasets or model versions. The final Task A and Task B notebooks are the source of truth for the completed handoff.

6. Suggested Use After Handoff

- Use the final notebooks as reproducible templates for retraining on updated data.
- Use the dataset creation workflow notebook to verify or recreate the validated CSV when raw DSSAT outputs change.
- Use the demo notebooks and HTML exports for communication with non-technical stakeholders.
- Recalibrate Task B thresholds if the model is retrained, if the dataset changes, or if the client decides to prioritize fewer missed transitions versus fewer false alerts.
- Keep appendix notebooks for traceability, but avoid using exploratory notebooks as the source for final reported numbers unless explicitly noted.